Solution Requirements

Date	15 July 2026
Team ID	SWUID2026217449
Project Name	Rising Waters - Flood Prediction ML
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	*No authentication required - open access web application*	Low
2	Authorization levels	*Single user access - no roles or permissions needed*	Low
3	External interfaces	*Kaggle HDI dataset used as external data source*	High

		services the system must connect to]	
4	Transactions processing	*User submits input form, model processes and returns HDI score*	High
5	Reporting	*Display HDI score and development category on result page*	High
6	Business rules, etc.	*HDI >= 0.8=Very High, 0.7-0.8=High, 0.55-0.7=Medium, <0.55=Low*	High
7	Compliance to laws or regulations	*Uses open source UNDP HDI dataset from Kaggle*	Medium
8	*Other*	*Model accuracy must be above 90% R2 score*	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	*Prediction response time under 2 seconds*	*Page loads in < 2 seconds*
2	Scalability	*Handles all 195 country HDI predictions accurately*	*Supports 195 countries dataset*
3	Security & Data Privacy	*No personal data collected or stored in the system*	*No PII stored, open access*
4	Reliability & Availability	*Application runs without crashes during normal usage*	*99% uptime on local server*
5	Usability & Accessibility	*Simple form-based interface, easy for any user to operate*	*User-friendly, no training needed*
6	*Other*	*ML model R2 score must be above 95% for reliable predictions*	R2 Score = 97.73%